

ECSE 4660/6660: Internetworking of Things

Lecture 5: PHY Layer Fundamentals – Part II

Ish Jain
ECSE, RPI

Spring 2026

Administrative

- Use Piazza for Lab questions
 - <https://piazza.com/rpi/spring2026/ecse46606660>
 - Asking Questions by email or Webex is not recommended
 - Piazza allows peer-discussion and faster response (can even post anonymously)
 - 5% Class Participation grades
- Today's Quiz is available on **LMS**
 - Multiple choice questions
 - Due tonight 11:59 PM
- Project group formation **due 1/30**
 - Add Project Group here [IoT_SP26_project.xlsx](#)
- Project Proposal submission **due 2/6**
 - Template Provided
 - Rubrics Provided
- HW1 to be released this Friday on **LMS** and due in 1 week: **due 2/6**
- Lab is located in JEC 5312
 - Access is provided to all registered students
 - Collect Lab-kit (Rpi etc) during TA hours
 - You can visit lab anytime Mon-Fri 7AM-10PM
 - Labs are done **individually**-Plagiarism penalized heavily (See Syllabus pdf)
 - **TA hours 4-5 PM** daily in 5312 lab
 - Lab 1 released; due **Fri, 1/30**
 - Lab 2 to be released this week; due **Fri, 2/13**

Proposal Template

[A descriptive title]

ECSE 4660/6660 Internetworking of Things

[Date]

[team member1, abcde@rpi.edu, ECSE 4660]

[team member2, abcde@rpi.edu, ECSE 4660]

1. Objective

This project will develop xx and demonstrate yy. xx will help people achieve goal zz. Our idea/ demonstration will benefit society/students/institutes/world because... (Think of this as your next **startup pitch**). This should identify clearly the **research question** you're addressing.

2. Innovation and Literature Review

This is an **innovative idea** and never demonstrated before in this form – some relevant projects found online are listed below... We are different from these works because...



3. Methodology

A clear statement of your **research methods**. i.e., how are you going to solve the problems you've raised and motivated in the previous paragraph?

We will utilize xyz sensors and develop a protocol to communicate the data over abc.

The system diagram is shown in Figure 1. (One or more representative figures or **block diagram** to show your hardware/system flow.)

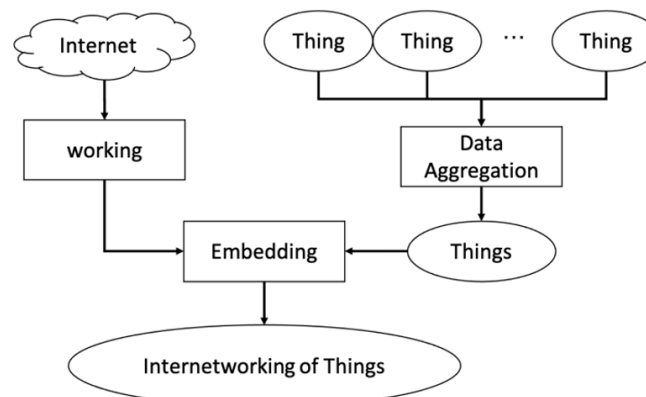


Figure 1 A schematic diagram of XX.

Proposal Template

4. Hardware/Software Used

We will use **compute platform**, e.g., RPi's, such-and-such **sensors**, PC running program V to visualize output, etc. The **program** will be written in Python/C/Java..., and the sensors will be from the IoT lab.

Make sure to check the list of sensors posted on Piazza. If the sensor is not listed and you don't possess them yourself, we might be able to order from Amazon up to certain budget. **Highlight** the need to buy new sensors/hardware in your report.

5. Demo and Deliverable

We will **demonstrate** the use and operation of XX in a **live demo** scenario. The algorithm/protocol will be shown by simulated/real application/or both. Students at 6000 level – describe your

security component here.

6. Plan and Timeline

A statement of plan and schedule, preferably in a table format, to convince us (and yourself!) that you can complete the project by the end of the term.

Actionable Item	Timeline
x	Mar 2 -16 (two weeks)
y	

Lab 1: Linux, Rpi, and Sensors

Part 1 — Linux Basics (VM + SSH/SCP)

- Set up Ubuntu VM (recommended prebuilt VirtualBox image)
- Practice core commands + package installs (apt update/upgrade)
- Install/enable SSH server, and use SSH + SCP
- Publish a small “bio” file to your RPI homepage server

Part 2 — Raspberry Pi + Sensors

- Configure the RPi (monitor setup OR headless SD-card setup)
- Enable SSH, connect from your laptop, transfer files
- Interface sensors (DHT11, HC-SR04) and capture evidence
- Build a simple RFID app (RC-522): write tag → read tag

Wireless Overview

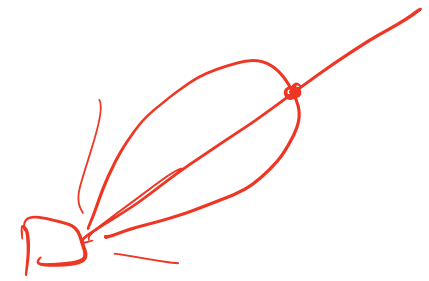
1. Overview of Wireless technologies
2. Antenna Design
3. Attenuation
4. Free space loss
5. Noise
6. Multipath & Fading (NLoS)

References

- V. Garg, *Wireless Communications & Networking*, Morgan Kauffman, 2007. (available as e-book for RPI students)
 - Chapter 3: Radio Propagation and Path Loss Models
 - Chapter 10: Antennas, Diversity, and Link Analysis
- L. Couch, *Digital and Analog Communication Systems*, Pearson, 2012.
 - Chapter 1: Introduction
 - Chapter 8: Wire and Wireless Communication Applications

Piazza reminder!

Antenna Gain



- Antenna gain: $G(\theta)$
 - Power output, in a particular direction, compared to that produced in any direction by a perfect omnidirectional antenna (isotropic antenna)
 - The increased power in a given direction is at the expense of other directions
 - Refers to directionality, i.e. relative amplification in a certain direction
- Effective area A_e
 - Related to physical size and shape of antenna

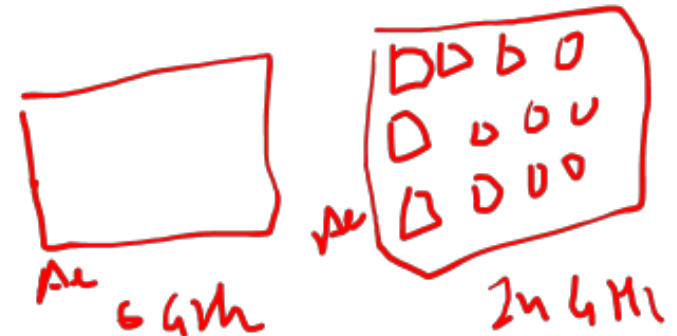
Antenna Gain (Cont'd)

- Relationship between antenna gain and effective area

$$G = \frac{4\pi A_e}{\lambda^2} = \frac{4\pi f^2 A_e}{c^2}$$

Handwritten annotations: A red box around G , a red arrow pointing from the box to the first equation, a red circle around A_e in the second equation, and two red arrows pointing upwards from the second equation, labeled f and G .

- G = antenna gain
- A_e = effective area
- f = carrier frequency
- c = speed of light ($= 3 \times 10^8$ m/s)
- λ = carrier wavelength
- Note: $\lambda f = c$



Antenna Gains and Effective Areas

TABLE 8-4 ANTENNA GAINS AND EFFECTIVE AREAS

Type of Antenna	Power Gain, G_A (absolute units)	Effective Area, A_e (m ²)
Isotropic	1	$\lambda^2/4\pi$
Infinitesimal <u>dipole</u> or loop	1.5	$1.5\lambda^2/4\pi$
Half-wave dipole	1.64	$1.64\lambda^2/4\pi$
Horn (optimized), mouth <u>area</u> , A	$10A/\lambda^2$	$0.81A$
Parabola or “ <u>dish</u> ” with <u>face area</u> , A	$7.0A/\lambda^2$	$0.56A$
Turnstile (two crossed dipoles fed 90° out of phase)	1.15	$1.15\lambda^2/4\pi$

- miniature antennas

[Src: Chapter 8, Couch]

Example

- Q: A parabolic reflective antenna with a diameter of 2m operating at 12 GHz. Find the antenna gain.

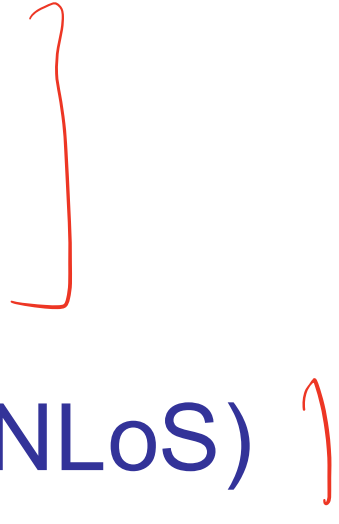
- G = antenna gain
- A_e = effective area
- f = carrier frequency
- c = speed of light ($= 3 \times 10^8$ m/s)
- λ = carrier wavelength

$$G = \frac{4\pi A_e}{\lambda^2} = \frac{4\pi f^2 A_e}{c^2}$$

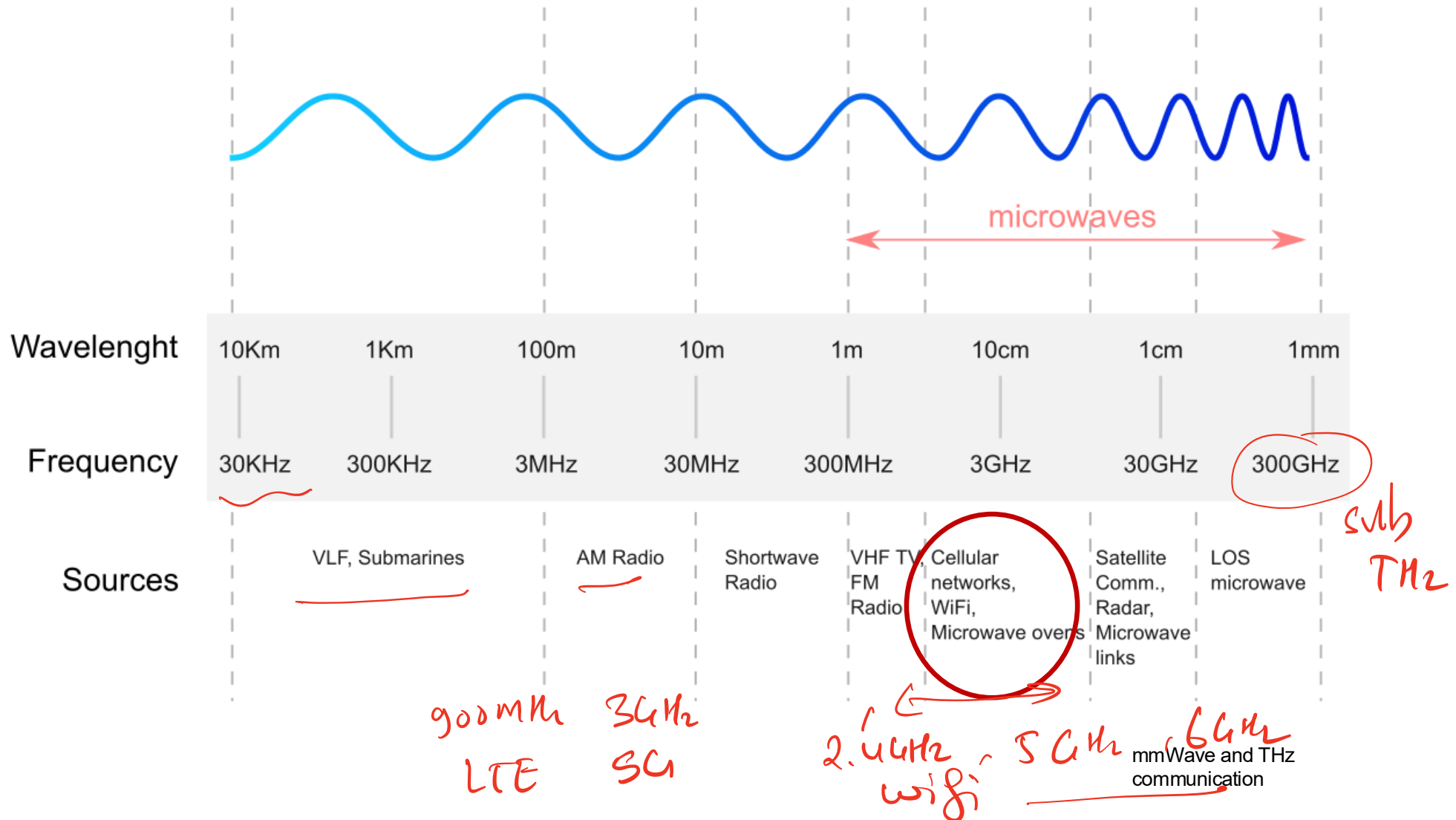
TABLE 8-4 ANTENNA GAINS AND EFFECTIVE AREAS

Type of Antenna	Power Gain, G_A (absolute units)	Effective Area, A_e (m ²)
Isotropic	1	$\lambda^2/4\pi$
Infinitesimal dipole or loop	1.5	$1.5\lambda^2/4\pi$
Half-wave dipole	1.64	$1.64\lambda^2/4\pi$
Horn (optimized), mouth area, A	$10A/\lambda^2$	$0.81A$
Parabola or “dish” with face area, A	$7.0A/\lambda^2$	$0.56A$
Turnstile (two crossed dipoles fed 90° out of phase)	1.15	$1.15\lambda^2/4\pi$

Wireless Overview

1. Overview of Wireless technologies
 2. Antenna Design
 3. **Attenuation**
 4. **Free space loss**
 5. **Noise**
 6. Multipath & Fading (NLoS)
- 

Radio Frequency Spectrum



1. Attenuation & attenuation distortion

- Strength of signal falls off with distance over transmission medium
- Attenuation factors for unguided media:
 - Received signal must have sufficient strength so that circuitry in the receiver can interpret the signal
 - Signal must maintain a level sufficiently higher than noise to be received without error
 - Attenuation is greater at higher frequencies, causing distortion
 - freq components of received signal have different relative attenuation
 - solution using “**equalizers**” that equalize the attenuations across the frequency band

physics
of
EM

‘link budget’

2. Free Space Loss

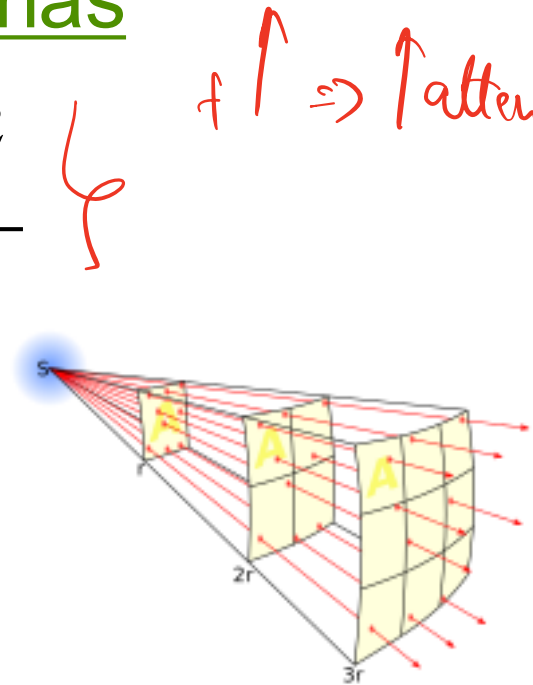
- Free space loss, ideal isotropic antennas

Path loss

$$\frac{P_t}{P_r} = \frac{(4\pi d)^2}{\lambda^2} = \frac{(4\pi f d)^2}{c^2}$$

- P_t = signal power at transmitting antenna
- P_r = signal power at receiving antenna
- λ = carrier wavelength
- d = propagation distance between antennas
- c = speed of light (3×10^8 m/s)

where d and λ are in the same units (e.g., meters)



$$d = \frac{c}{f}$$

$$P_r = \left(\frac{c^2}{(4\pi f d)^2} \right) P_t$$

Free Space Loss

- Free space loss equation can be recast:

$$10 \log_{10}(\text{loss})$$

$$\underline{L_{dB}} = 10 \log \frac{P_t}{P_r} = \underline{20 \log} \left(\frac{4\pi d}{\lambda} \right)$$

$$20 \log_{10}(4\pi)$$

$$= \underline{-20 \log(\lambda)} + \underline{20 \log(d)} + \underline{21.98 \text{ dB}}$$

$$= 20 \log \left(\frac{4\pi f d}{c} \right) = \underline{20 \log(f)} + \underline{20 \log(d)} - \underline{147.56 \text{ dB}}$$

rewrite
in
terms of
freq

$$10 \log 12
10 e^3$$

unit less

Understanding ^{dB,} dBW and dBm, ^{dB_i}

define dBW = $10 \log_{10} (\text{Power in watt})$

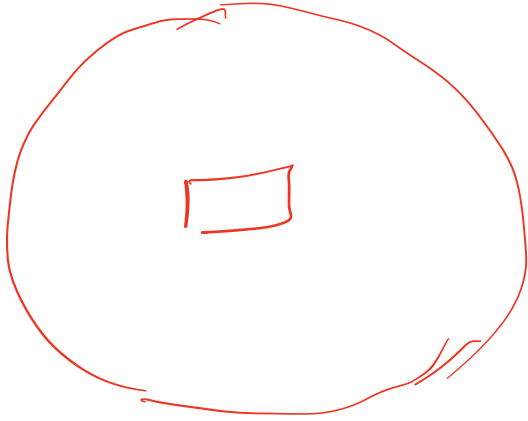
dBm = $10 \log_{10} (\text{Power in milli watt})$

Transmit power = $P_t (w)$

in dBW $\Rightarrow 10 \log_{10} \left(\frac{P_t}{1w} \right)$

in dBm $\Rightarrow 10 \log_{10} \left(\frac{P_t (w)}{1mw} \right)$

what is dBi
⇒ Power relative to isotropic antenna



Directional antenna



G in dBi :

Power or gain

$$10 \log_{10} \left(\frac{G(\omega)}{\underbrace{G_{\text{iso.}}(\omega)}_{= 1}} \right)$$

unitless

Free Space Loss

- Free space loss accounting for gain of other antennas

$$\text{Loss} = \frac{P_t}{P_r} = \frac{(4\pi)^2 (d)^2}{G_r G_t \lambda^2} = \frac{(\lambda d)^2}{A_r A_t} = \frac{(cd)^2}{f^2 A_r A_t}$$

- G_t = gain of transmitting antenna
- G_r = gain of receiving antenna
- A_t = effective area of transmitting antenna
- A_r = effective area of receiving antenna

$$\text{Recall that: } G = \frac{4\pi A_e}{\lambda^2} = \frac{4\pi f^2 A_e}{c^2}$$

Free Space Loss

- Free space loss accounting for gain of other antennas can be recast as

$$\begin{aligned} \underline{L_{dB}} &= 20 \log(\lambda) + 20 \log(d) - 10 \log(A_t A_r) \\ &= -20 \log(f) + 20 \log(d) - 10 \log(\underline{A_t A_r}) + 169.54 \text{dB} \end{aligned}$$

Example 1 (in Quiz today)

- Find the received power for the link from a synchronous satellite to a terrestrial antenna.

Use the following data: Height: 80,000 km.

Satellite transmitted power: 8W. Transmit antenna gain: 20 dBi. Receive antenna gain: 60 dBi. Transmit frequency: 20 GHz

$d = 80 \times 10^6 \text{ m}$

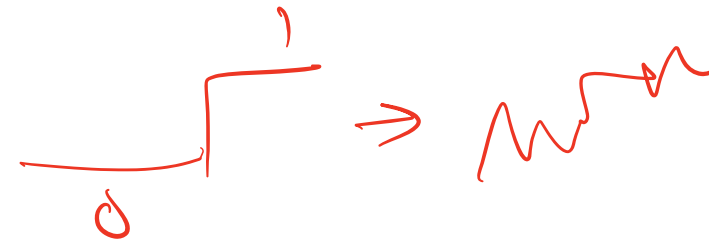
P_t

G_r

f

P_r

Categories of Noise



- Noise is a major limiting factor in communication system performance
- Unwanted signals in the medium are added to the intended signal at the receiver

- Categories of noise:

- Thermal Noise
- Intermodulation noise
- Crosstalk
- Impulse Noise

temperature-dependent

3. Thermal Noise

- Thermal noise due to agitation of electrons
- Present in all electronic devices and transmission media
- Cannot be eliminated x
- Function of temperature
- Particularly significant for satellite communication , low power IoT

long. range

Rx has more noise

Thermal Noise

- Amount of thermal noise to be found in a bandwidth of 1Hz in any device or conductor is:

$$N_0 = kT \text{ (W/Hz)}$$

Handwritten notes: 'Joule' with arrows pointing to 'k' and 'T'.

- N_0 = noise power density in watts per 1 Hz of bandwidth
- k = Boltzmann's constant = 1.3803×10^{-23} J/K *6*
- T = temperature, in kelvins (absolute temperature)

$H_2 = \frac{1}{\text{sec}}$

total noise $B = 20 \text{ mHz}$

$N = kT \times B \text{ Watt}$

Thermal Noise



- Noise is assumed to be independent of frequency
- Thermal noise (in watts) present in a bandwidth of B Hertz:

$$\underline{N = kTB} \quad \text{watt} \quad \sim \quad E(n^2)$$

or, in decibel-watts

$$\begin{aligned} \underline{N} &= 10 \log k + 10 \log T + 10 \log B \\ &= \underbrace{(-228.6)}_{\text{dBW}} + 10 \log T + 10 \log B \quad \text{dBW} \end{aligned}$$

Signal-to-Noise Ratio

- Ratio of the power in a signal to the power contained in the noise that's present at a particular point in the transmission
- Typically measured at a receiver
- Signal-to-noise ratio (SNR, or S/N)

$$(\text{SNR})_{\text{dB}} = 10 \log_{10} \left\{ \frac{\text{signal power}}{\text{noise power}} \right\}$$

at receiver

w

w

- A high SNR means a high-quality signal, low number of required intermediate repeaters
- SNR sets upper bound on achievable data rate

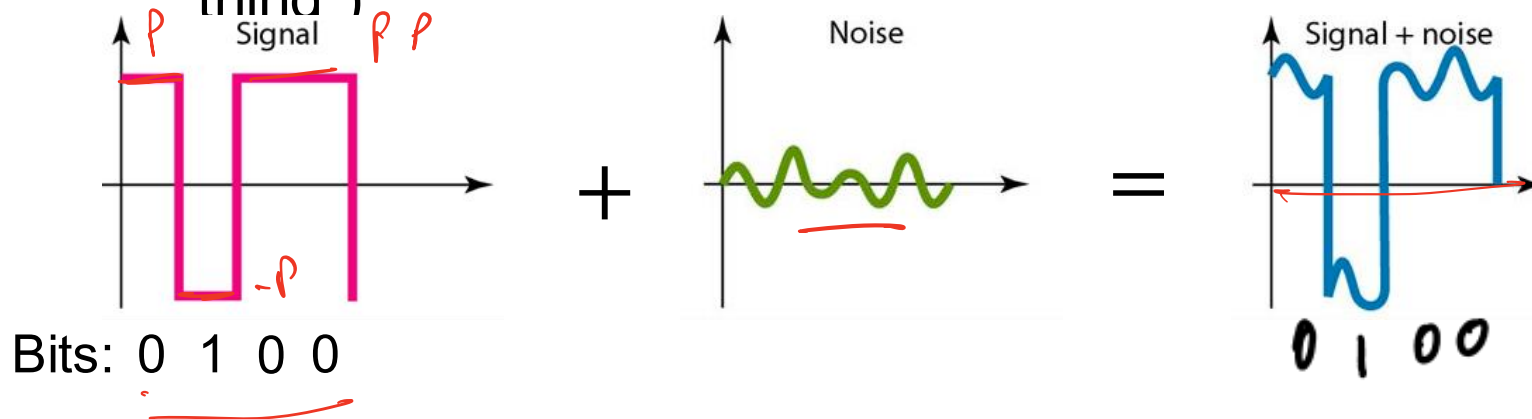
noise floor = min power required at Rx for viable comm.

SNR: Signal-To-Noise Ratio

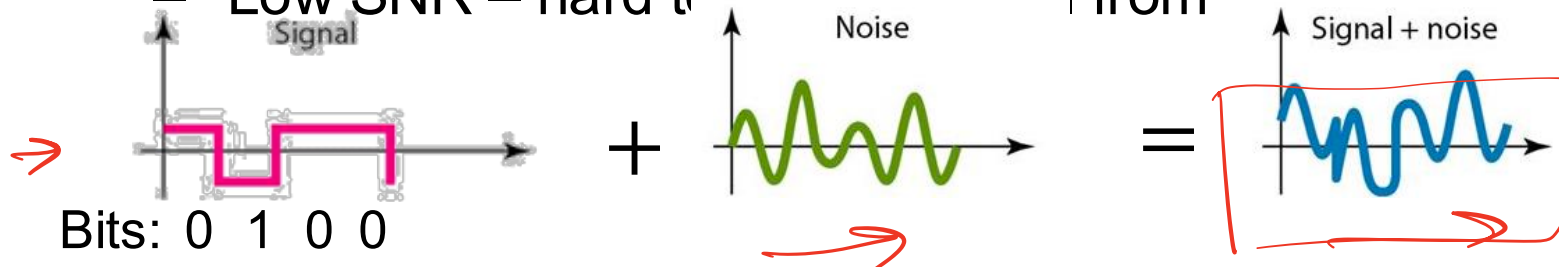
$$SNR = \frac{\text{Received Signal Power}}{\text{Noise Power at RX}}$$

TX power
wirechannel
Hardware

- High SNR – easier to extract signal from noise (a “good thing”)



- Low SNR – hard to extract signal from noise (a “bad thing”)



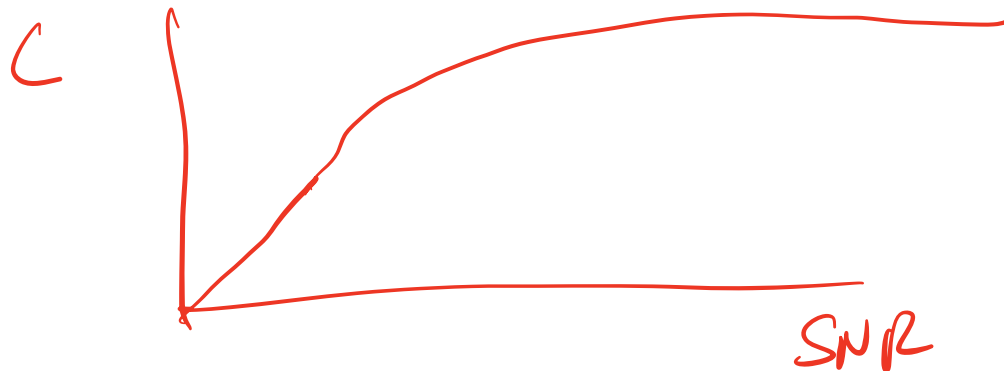
Shannon Capacity

- The maximum channel capacity in bits per second is

$$C = B \log_2(1 + \text{SNR})$$

where B is the channel bandwidth in Hz.

This result assumes white (Gaussian) noise, no attenuation or delay distortion.



$\frac{P_r}{N}$ theoretical max limit

Example 2 (in Quiz today)

- Given a channel with intended capacity of 90 Mbps, the bandwidth of the channel is 8 MHz. What signal-to-noise ratio is required to achieve this capacity?

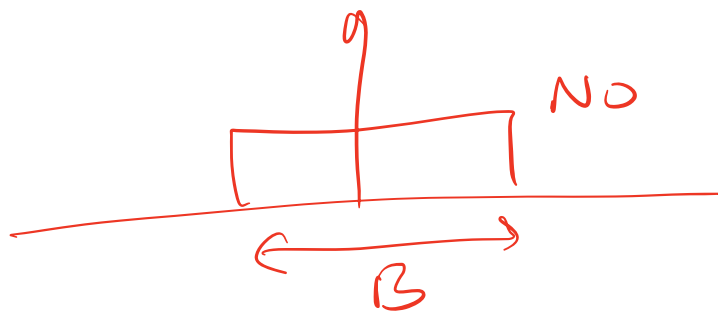
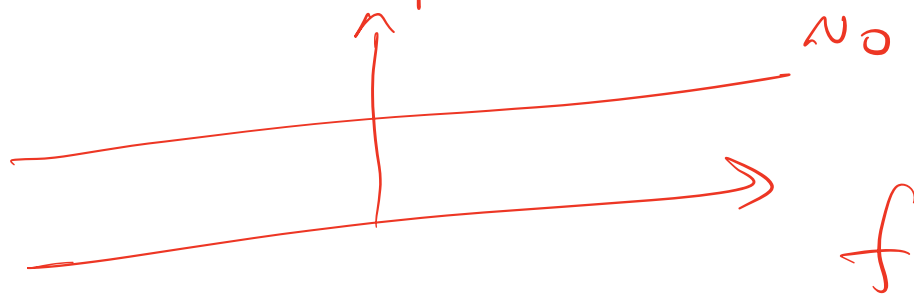
Intermodulation noise

- may happen when the Tx, Rx or medium behave in non-linear ways
- when signals at different frequencies share the same medium, **non-linearities cause signals at sum or difference of original frequencies**
- these can interfere with legitimate signals at those frequencies



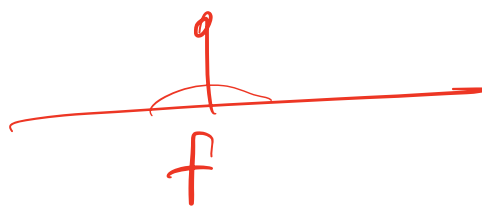
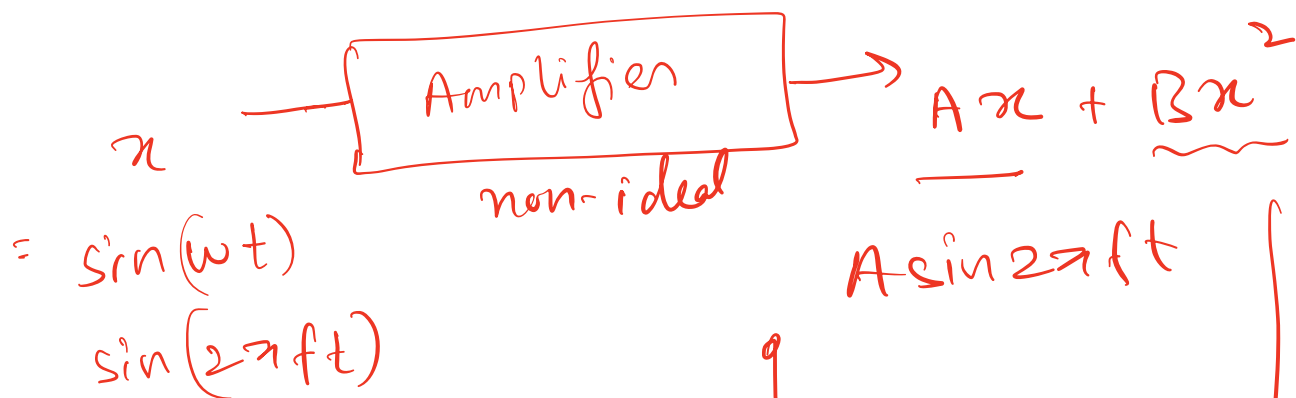
$$y = h x + n$$

(1) Thermal noise power density



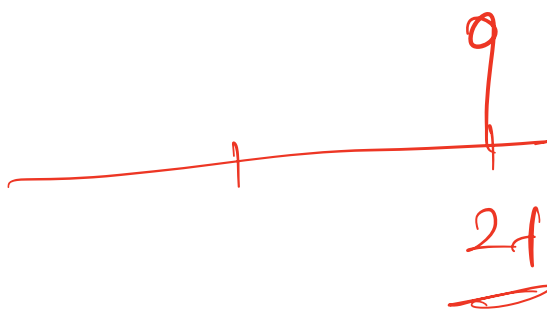
$\Rightarrow$ total noise
 $N = N_0 B$

(2) inter modulation noise



$B(\sin 2\pi ft)^2$

$$\frac{(1 - \cos(2\pi \cdot 2ft))}{2}$$



Crosstalk

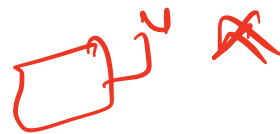
wired

- unwanted coupling between signal paths
- e.g. crosstalk in phone conversations

interference



$$y = hx + \alpha + \text{noise}$$



Impulse Noise

- noise spikes of short duration and high magnitude
- may happen due to lightning or other electromagnetic disturbances



Wireless Overview

1. Overview of Wireless technologies
2. Antenna Design ✓
3. Attenuation ✓
4. Free space loss ✓
5. Noise ✓
6. **Multipath & Fading (NLoS)**

non line of sight

Multipath Propagation

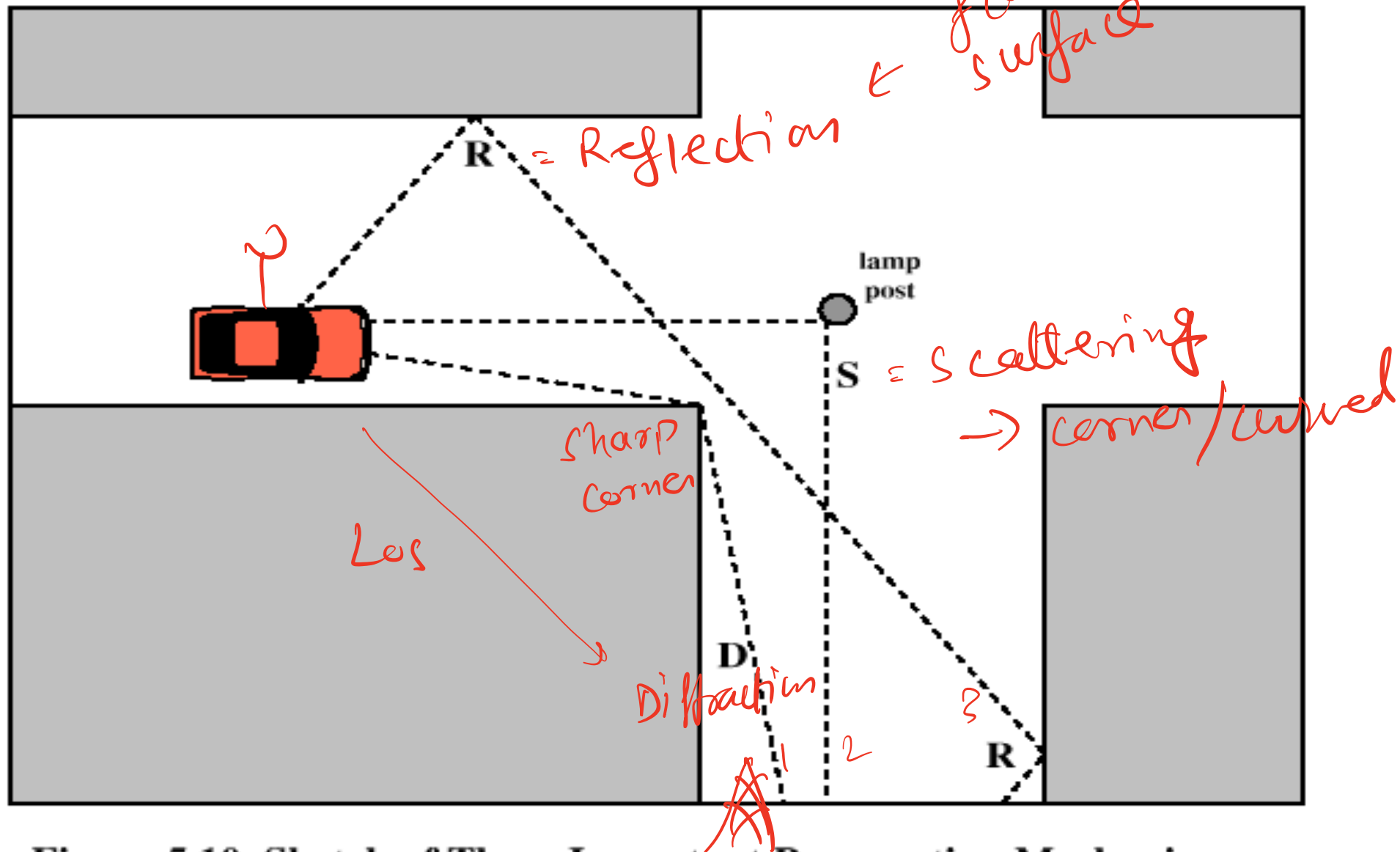


Figure 5.10 Sketch of Three Important Propagation Mechanisms: Reflection (R), Scattering (S), Diffraction (D) [ANDE95]

Multipath Propagation

$\uparrow$ than λ

- **Reflection** - occurs when signal encounters a surface that is large relative to the wavelength of the signal
- **Diffraction** - occurs at the edge of an impenetrable body that is large compared to wavelength of radio wave
- **Scattering** - occurs when incoming signal hits an object whose size is in the order of the wavelength of the signal or less

$\approx$ to λ

The Effects of Multipath Propagation

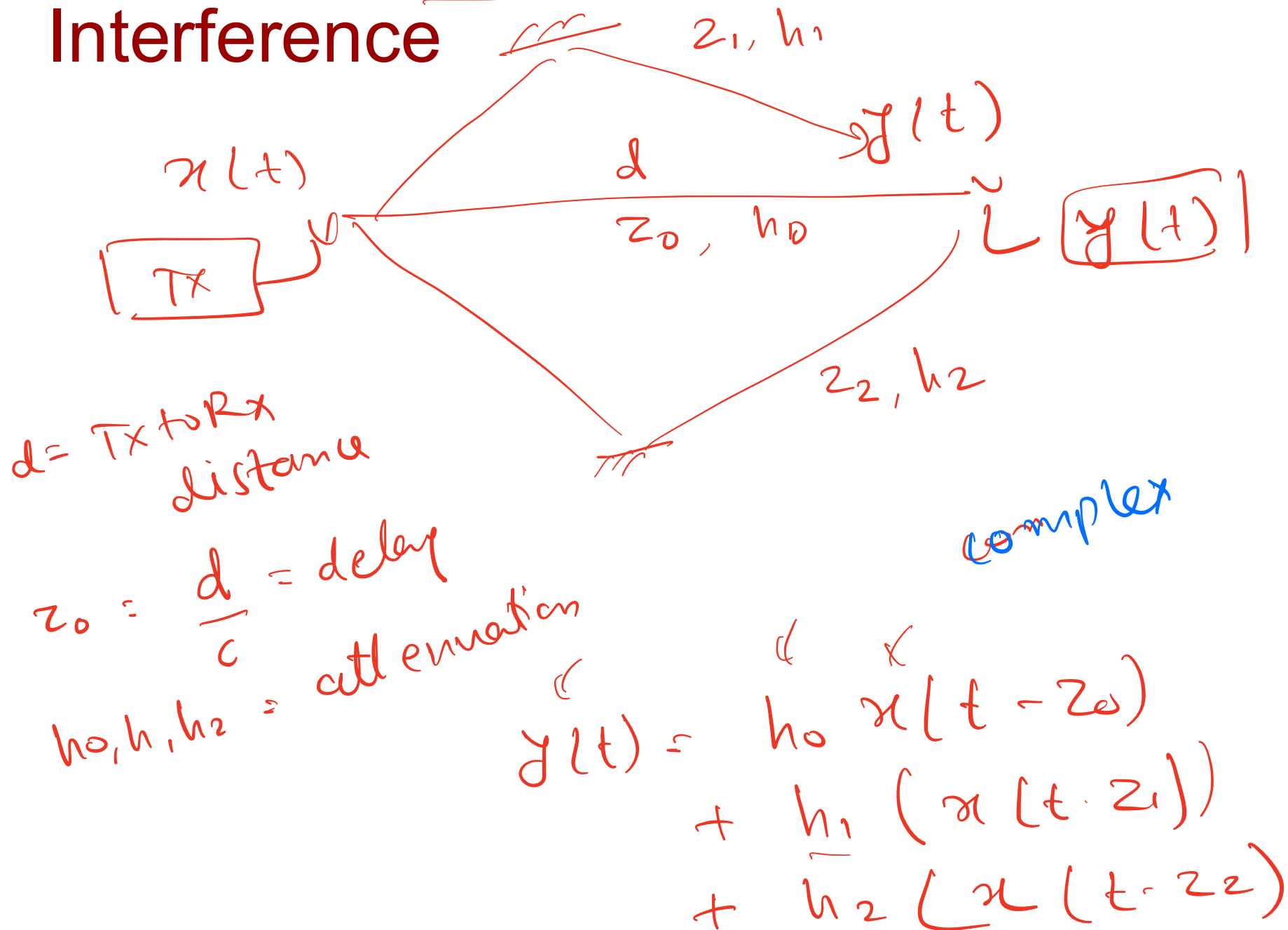
■ Fading

- Multiple copies of a signal may arrive at different phases
- If phases add destructively, the signal level relative to noise declines, making detection more difficult }

■ Inter-Symbol Interference (ISI)

- One or more delayed copies of a pulse may arrive at the same time as the primary pulse *for a subsequent bit*

Fading – constructive and destructive Interference

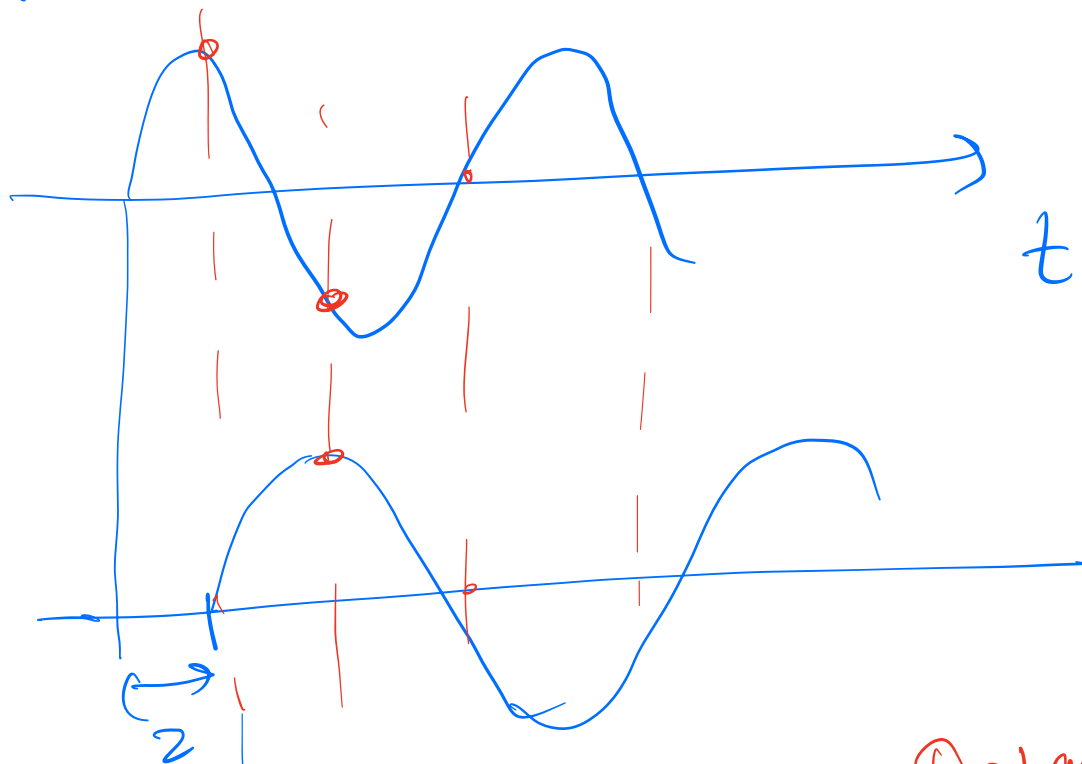


$$x(t) = \cos 2\pi f t$$

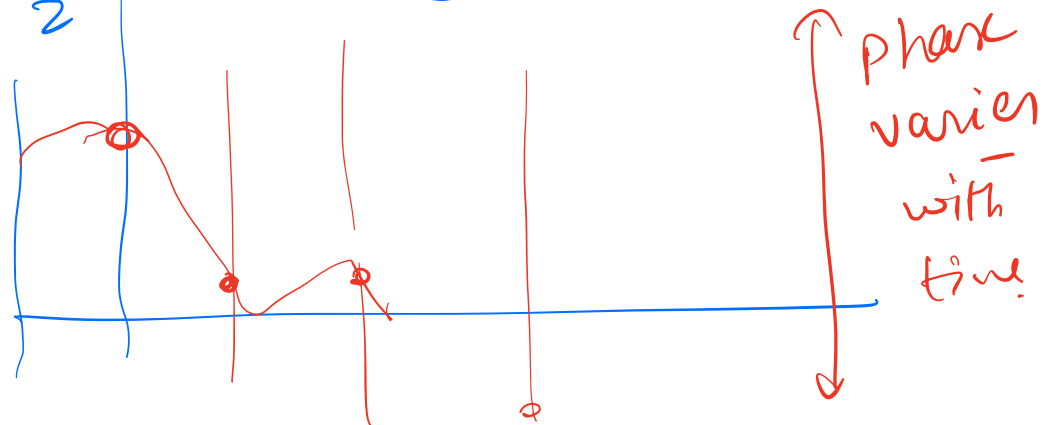
$$y(t) = x(t) + x(t-2)$$

$$= \cos 2\pi f t + \cos 2\pi f (t-2)$$

$\cos 2\pi f t$



$\cos 2\pi f t + \cos 2\pi f (t-2)$



$\Rightarrow$ FFT $\Rightarrow$ look for freq domain

Inter-Symbol Interference – time domain

“Fading”

- Fading is the change in received power of a signal over time due to multi-path.
- Types of Fading
 - Slow fading
 - Fast fading
 - Rayleigh fading
 - Rician fading
 - Flat fading
 - Frequency-Selective fading

Slow vs Fast Fading

- In urban (or non free-space) environments, the two common kinds of 'fading' may take place
 - **Slow fading**: typically due to the movement of obstacles or transmitter/receiver, causing slow variation in the signal strength
 - **Fast Fading**: due to multipath propagation

Shadow (slow) fading

- The received signal strength from the transmitter will be different – i.e. it will vary around the mean value given by the path-loss equation.
- It is called 'slow' because the variations are typically much slower than 'fast fading'
- Several measurements and simulations indicate that this variation can be expressed as a log-normal random variable
- It is typically compensated for by increasing the transmit power by an amount to guarantee a certain 'fading margin'

Shadow (slow) fading

Multi-path (fast) fading

- A signal may be received at the receiver through different 'paths' creating copies with different phase.
- Fast fading is the fluctuation of the signal amplitude because of the addition of signals arriving with different 'phases'
- The phase of the arriving paths changes rapidly, causing fast fluctuations of the signal amplitude
- The distribution of the amplitude follows:
 - Rayleigh distribution: if all signals suffer nearly the same attenuation, but arrive with different phases, and no distinct dominant path (outdoor)
 - Rician distribution: when there is a strong LOS signal component (indoor)
- Mitigation:
 - Increasing signal power is not very helpful
 - Error control coding with interleaving
 - Diversity schemes
 - Directional Antennas (reducing the number of multiple paths)

Multi-path (fast) fading – Intuition

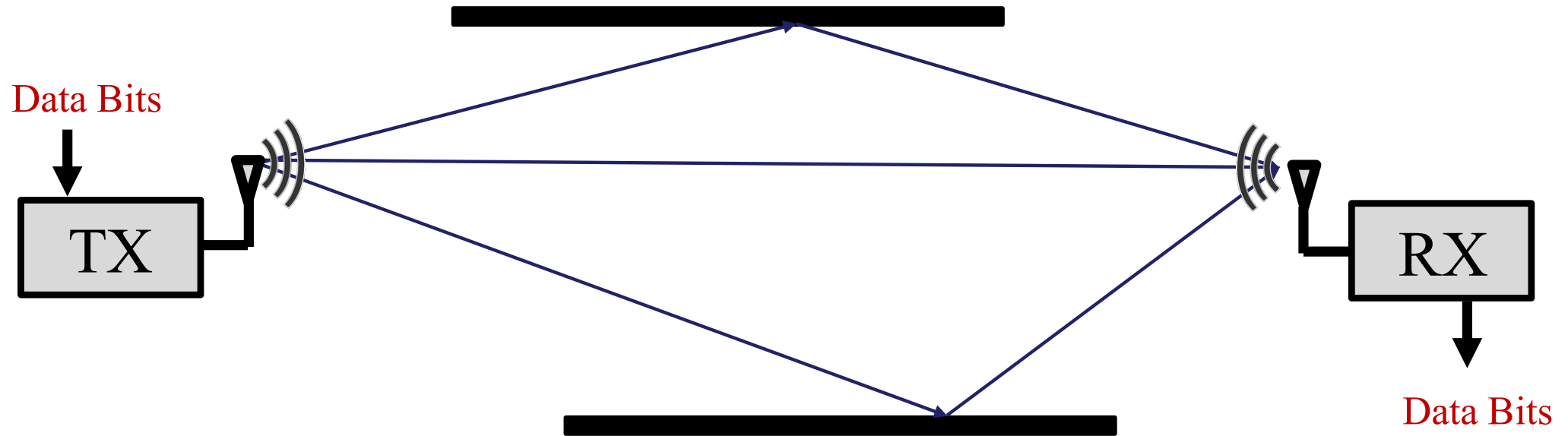
Multipath Delay Spread (ISI)

- If the multiple paths also introduce **delays** that are comparable to (or larger than) the symbol duration, the received waveform **spreads** into the neighboring symbols and produces ISI (irreducible error rates).
- Several models exist for multipath delay spread
- Mitigation: (we will study these later)
 - Equalization
 - Direct Sequence Spread Spectrum
 - Directional Antennas (reducing the multipath components)

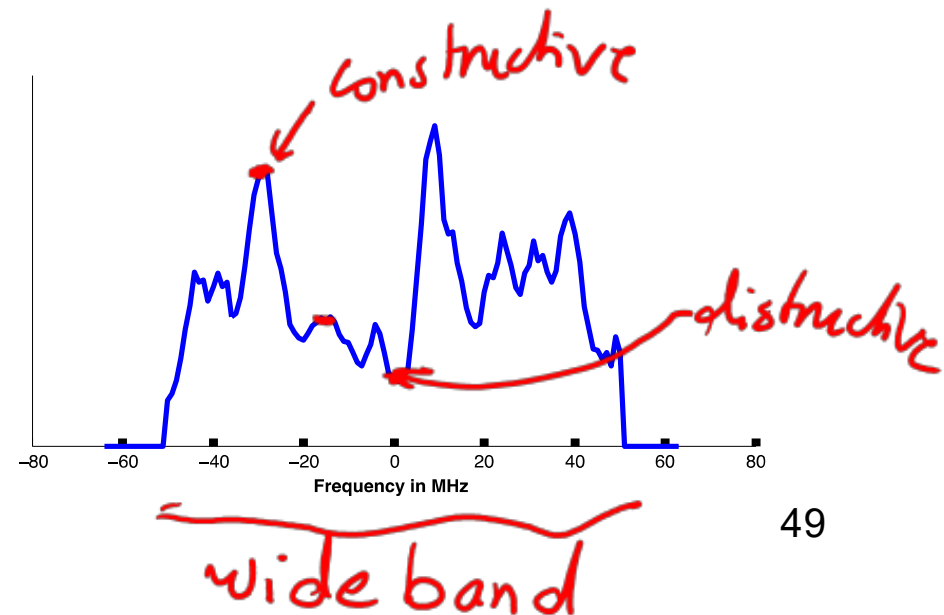
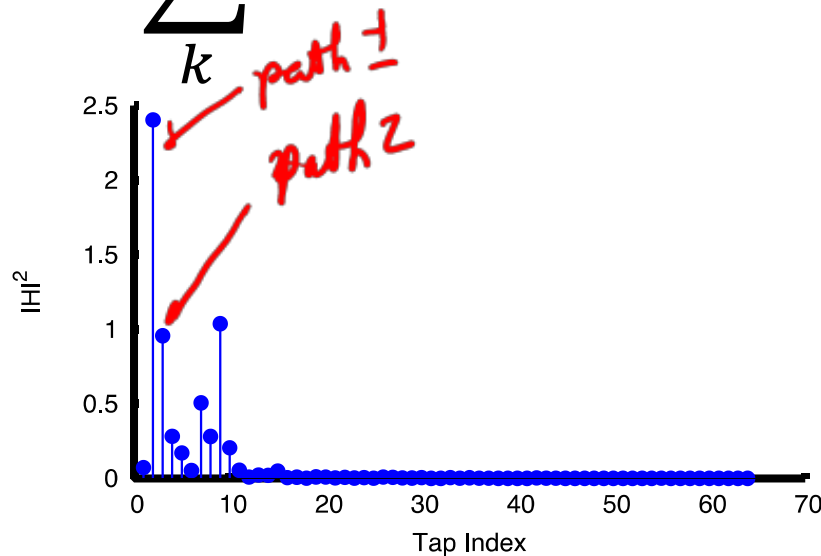
ISI – Frequency Domain Interpretation (Flat vs Frequency-selective Fading)

ISI – Frequency Domain Interpretation (Flat vs Frequency-selective Fading)

Wide Band vs Narrow Band Channel

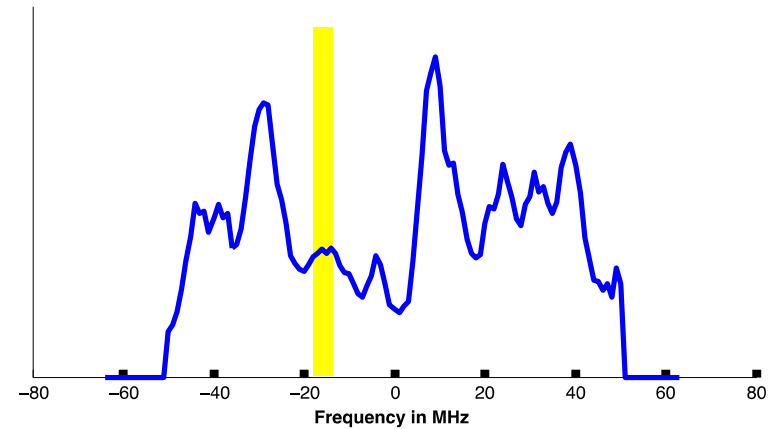
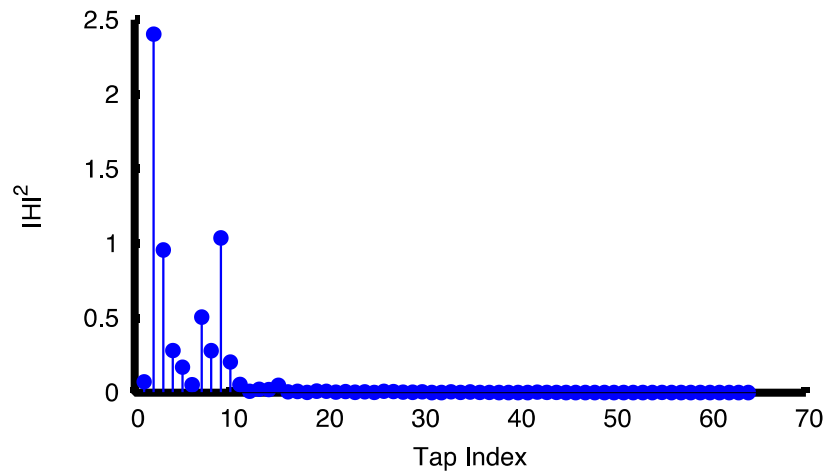


$$y(t) = \sum_k h_k x(t - \tau_k) = \underline{h(t)} * x(t) \Leftrightarrow H(f)X(f)$$



Wide Band vs Narrow Band Channel

$$y(t) = \sum_k h_k x(t - \tau_k) = h(t) * x(t) \Leftrightarrow H(f)X(f)$$



Wide Band vs Narrow Band Channel

$$y(t) = \sum_k h_k x(t - \tau_k) = h(t) * x(t) \Leftrightarrow H(f)X(f)$$

